

Shadow Price Challenge

Consider the linear program

$$\max_{x_1, x_2} \quad z = \pi_1 x_1 + \pi_2 x_2$$

subject to

$$a_{11}x_1 + a_{12}x_2 \leq b_1$$

$$a_{21}x_1 + a_{22}x_2 \leq b_2$$

$$a_{31}x_1 + a_{32}x_2 \leq b_3$$

$$x_1, x_2 \geq 0.$$

Recall that the shadow price of resource j is the marginal change in the optimized objective value from a small increase in b_j :

$$\lambda_j = \frac{\partial z^*}{\partial b_j}.$$

Sam proposes the following shortcut:

$$\lambda_j = \frac{\pi_i}{a_{ji}}$$

That is, “the shadow price of a constraint is just the objective-function coefficient of a variable divided by the amount of that resource required to produce one unit of the variable.”

Your task: Determine whether this statement is always true.

If it is not always true, identify a set of conditions under which it is true. Your answer should include a mathematical argument, not just a numerical example.

As you work, consider the following questions:

- What must be true about constraint j at the optimum for it to have a nonzero shadow price?
- If b_j increases slightly, which decision variable or variables change?
- What would have to be true for

$$\frac{\partial x_i}{\partial b_j} = \frac{1}{a_{ji}}?$$

- What could cause the shortcut to fail even if constraint j is binding?
- Does your argument apply to arbitrarily large changes in b_j , or only to sufficiently small changes?

Challenge: Draw a graph of a two-variable LP for which the shortcut works, and then modify the problem so that the same shortcut fails.